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An Artificial Bee Colony Based PID Tuning Algorithm for Inverted Pendulum and Cart System

Sabir Husnain
Dept. of Mechanical Mechatronics and Manufacturing Engineering, UET Lahore, Faisalabad Campus
Faisalabad, Pakistan
sabirhasnain577@gmail.com

Muhammad Tanveer Riaz
Dept. of Mechanical Mechatronics and Manufacturing Engineering, UET Lahore, Faisalabad Campus
Faisalabad, Pakistan
tanveer.riaz@ieee.org

Muhammad Abbas Khan
Dept. of Electrical Engineering Balochistan university of information technology engineering and management sciences
Quetta, Pakistan
Muhammad.Abbas@buitms.edu.pk

Mahnoor Mazhar
Dept. of Chemistry
Government College Women University Faisalabad
Faisalabad, Pakistan
mahnoormazhar50@gmail.com

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***Abstract*—The Proportional, Integral, Derivative (PID) controller finds widespread use in various control applications, yet accurately tuning it can prove challenging and time-consuming, especially for intricate systems characterized by non-linearities and uncertainties. To address these challenges, an approach rooted in swarm intelligence-inspired optimization techniques, drawn from honeybee foraging behavior, is proposed for PID tuning, specifically employing the Artificial Bee Colony (ABC) algorithm. This algorithm is tailored for the control of an inverted pendulum and cart system. The ABC algorithm offers an efficient and automated avenue to navigate the parameter space and dynamically adjust PID gains according to performance criteria. Within this study, the inverted pendulum and cart model serves as a testing ground to evaluate the effectiveness of the proposed ABC-based PID tuning algorithm. The intricate presence of non-linearities and uncertainties in the system renders achieving precise control a formidable task. By harnessing the potential of the ABC algorithm, the objective is to uncover optimal PID gains that augment control performance, stability, and robustness amidst these complexities. The outcomes underscore the algorithm's adeptness in exploring the parameter space, tailoring gains based on performance feedback, and adeptly mitigating the system's non-linearities and uncertainties. The proposed approach bears multiple advantages, including diminished reliance on manual trial-and-error tuning, heightened automation, and improved control performance. By capitalizing on the strengths inherent to the ABC algorithm, engineers can streamline the PID tuning process and realize enhanced control outcomes for intricate systems.**

Keywords—Artificial Bee Colony (ABC), PID Control, PID Tuning, Inverted Pendulum and Cart, Artificial Bee Colony PID, Evolutionary Control Algorithm

I. Introduction

Control systems play a pivotal role in engineering by enabling the precise regulation and manipulation of physical processes. These systems utilize mathematical models, sensors, actuators, and mostly feedback loops to manage and maintain desired outputs in dynamic systems [1,2]. A widely known and extensively utilized control system in various industries is the PID (Proportional-Integral-Derivative) control system. This feedback control mechanism is renowned for its simplicity, effectiveness, and versatility in regulating dynamic processes. Tuning a PID controller for a given model can be a challenging task due to the complexity and nonlinearity of real-world systems [3]. The difficulty arises from the need to strike a delicate balance between several key parameters: the proportional gain (Kp), the integral gain (Ki), and the derivative gain (Kd). Each of these gains affects the system's response differently, and finding the optimal values requires a comprehensive understanding of the system dynamics and performance requirements.

PID tuning poses several challenges due to the complexity and variability of real-world systems. The inherent complexity of these systems, with their nonlinearities, dynamics, and interactions, makes it difficult to accurately model and tune the PID controller. Uncertainties and variability further complicate the process, as systems may exhibit parameter uncertainties, environmental variations, and disturbances that affect controller performance [3]. Balancing multiple performance criteria, such as stability, responsiveness, and robustness, presents a challenge, as optimizing one aspect may require trade-offs in others.

Evolutionary algorithms provide a powerful and efficient approach to simplify and optimize PID tuning. Evolutionary algorithms like Artificial Bee Colony (ABC) utilize a population-based search method inspired by natural evolutionary processes [4]. They explore the parameter space by iteratively generating and evolving potential solutions. Through the evaluation of these solutions based on performance criteria, the algorithm adaptively adjusts the PID gains to improve control performance [5,6].

This paper presents a study on PID tuning using the Artificial Bee Colony (ABC) algorithm, focusing on its application to an inverted pendulum and cart model. The inverted pendulum and cart model are used as test systems to validate the effectiveness of the control and ABC algorithm. By implementing the ABC algorithm for PID tuning, the paper demonstrates how this nature-inspired optimization technique can optimize the PID gains to achieve stable and accurate control of the inverted pendulum and cart system. The results

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obtained from the testing and validation of the control and ABC algorithm on the inverted pendulum and cart model provide evidence of the efficacy of using the ABC algorithm for PID tuning in real-world control applications.

The objectives of the paper, such as evaluating the performance of the ABC algorithm for PID tuning in the context of the inverted pendulum and cart model, specify the goals of the study, such as achieving stability, accuracy, and robustness in control, while minimizing steady-state error, input signal provided to the plant, and settling time.

II. Background Review

A. Artificial Bee Colony (ABC)

The Artificial Bee Colony (ABC) algorithm is a population-based optimization technique inspired by the foraging behavior of honey bees. It simulates the process of food source exploitation and recruitment within a bee colony. The primary goal of the ABC algorithm is to discover the optimal solution to an optimization problem through iterative exploration of the solution space using a population of artificial bees [7]. The algorithm commences by initializing a population of employed bees, with each bee representing a potential solution. These employed bees then explore the solution space by adjusting their positions based on the quality of their current food sources [8]. The quality of these sources is determined by evaluating the fitness function associated with each solution.

```
Initialize:
  Set population size (N)
  Set maximum number of iterations (MaxIter)
  Initialize population of employed bees with random solutions
Repeat until termination condition is met:
  Employed Bee Phase:
    for each employed bee:
      Generate a new solution by adjusting the current position
      Evaluate the fitness value of the new solution
  Onlooker Bee Phase:
    Calculate the probability of each employed bee's solution
    for each onlooker bee:
      Select a food source based on the probabilities
      Generate a new solution around the selected food source
      Evaluate the fitness value of the new solution
  Waggle Dance:
    Share information of fitness values of employed bees' solutions
  Scout Bee Phase:
    for each employed bee:
      if the solution does not improve after a number of iterations:
        Generate a new random solution
        Evaluate the fitness value of the new solution
  Update the best solution found so far
Return the best solution found
```

In each iteration, employed bees perform a local search around their current positions to discover new food sources. This search is guided by both their personal experience and information shared by onlooker bees [5]. Onlooker bees are responsible for selecting promising food sources based on their quality. The probability of selecting a particular food source is determined by its fitness value to other food sources. Onlooker bees choose food sources using a process similar to roulette wheel selection. The employed and onlooker bees collaborate by exchanging information through a process called the "waggle dance" [8]. This communication enables the onlooker bees to update their knowledge about the quality of different food sources, enhancing the exploration and exploitation capabilities of the algorithm. To maintain diversity in the search process, scout bees are introduced [10]. If a food source does not show improvement after a certain number of iterations, a scout bee abandons it and randomly explores a new solution. This helps prevent premature convergence and encourages exploration of different regions within the solution space.

Continue the iterations until a termination condition is met, such as reaching a maximum number of iterations or achieving a satisfactory solution based on a predefined criterion.

B. PID Control

The PID controller is extensively utilized in controlling various systems, including industrial applications such as motor speed control and temperature regulation. In this research, an inverted pendulum and cart system are used to demonstrate the PID control with the ABC algorithm [8]. This control system involves the task of balancing an inverted pendulum on top of a moving cart. The PID controller utilizes three key components: proportional, integral, and derivative terms. The proportional gain adjusts the control response proportionally and quickly to deviations, enhancing sensitivity. However, relying solely on the proportional gain may result in inaccuracies and steady-state errors [1, 3]. The integral gain addresses these issues by continuously integrating the deviation, effectively eliminating steady-state errors. Additionally, the derivative gain improves the system's response by anticipating and adjusting to changes in the system's dynamics, thus reducing overshoot and oscillation. Together, these three terms enable the PID controller to effectively stabilize and control the inverted pendulum and cart system, ensuring precise and balanced operation.

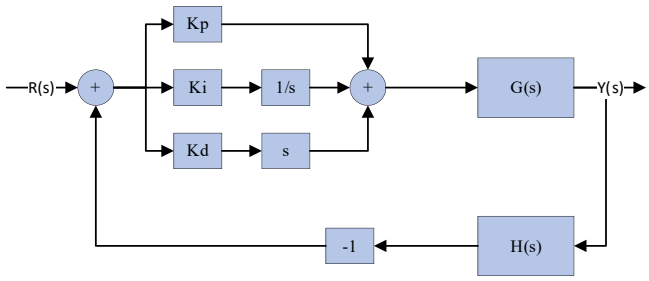

Fig. 1. Block diagram of PID controller

From figure 1, the mathematical representation of the PID controller in the form of a time domain can be shown in (1). The Laplace transform of the (1) can be written as (2) which can be stated as the transfer function of the system.

$$u(t) = K_p + K_i \int e.d(t) + K_d \dot{e} \tag{1}$$

$$U(s) = K_p + \frac{K_i}{s} + K_d s \tag{2}$$

III. Methodology

A. System Modelling

The inverted pendulum and cart system being presented in this paper is shown in figure 2 which can be used to develop a mathematical model of the system. The model consists of two major subsystems the electrical and the mechanical system. So, the system can be decomposed into these two subsystems [10,11]. The mechanical model of the inverted pendulum and cart can be represented using (3) and (4).

$$F = (M + m)\ddot{x} - ml\ddot{\theta} \tag{3}$$

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$$-mgl\theta = ml\ddot{x} - (I + ml^2)\ddot{\theta} \tag{4}$$

Where F is the force applied on the cart as an input by the actuator or external environment, x is the translational position of the cart, θ is the angular position of the pendulum, l is the half of the length of the pendulum rod or the position of its center of mass from its base, and the I is the moment of inertia of the pendulum.

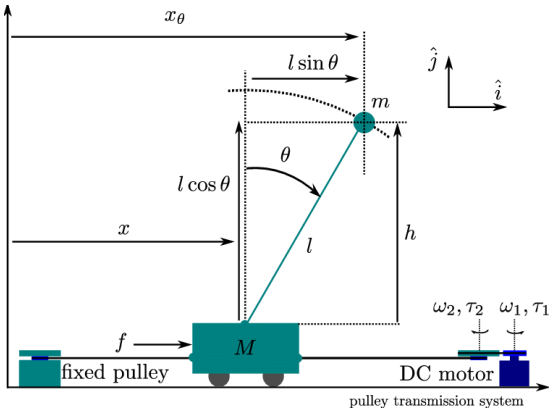

Fig. 2. Inverted Pendulum System

Now the electrical model can be represented using equations of DC motors (5) and (6), and then the two subsystem models can be joined together to get a complete model of the system that is an inverted pendulum and cart with DC motor [12].

$$T_m = K_m i = J_m\ddot{\theta}_m + b_m\dot{\theta}_m + T_L \tag{5}$$

$$\frac{L_m d(i)}{dt} + R_m i + K_m\dot{\theta}_m = V \tag{6}$$

Here the K_m is the motor constant, J_m is the inertia of the rotor of the motor, θ_m is the position of the motor, i is the motor current, R is the resistance of the armature, L is the inductance of the armature, and V is the input voltage of the motor.

The DC motor can again be split up into two components one is electrical and the other is mechanical as it is also an electromechanical system. It can also be stated that the electrical part of the motor has a more fast response than that of the mechanical part [13]. The dominant poles are only the poles of the mechanical transfer function; thus, the electrical part can be ignored for an approximated motor model which can be represented in first order using equations [13,14,16]. The approximated first-order system of the DC motor can be represented in (7) and (8) where $L_m \approx 0$.

$$R_m i + K_m\dot{\theta}_m = V \tag{7}$$

$$K_m i = J_m\ddot{\theta}_m + b_m\dot{\theta}_m + T_L \tag{8}$$

One can relate the motor position θ with the position of the cart x, using a simple relation that is $x = r\theta$, where r is the radius of the pullies attached to the motor and then with the cart using timing belt [18]. Thus, the equation can then be related with the force using relation $T_L = rF$. By putting value of T_L in (8) one can obtain (9).

$$F = \frac{K_m}{r}V - \frac{K_m}{r^2R_m}\dot{x} - \frac{J_m}{r^2}\ddot{x} - \frac{b_m}{r^2}\dot{x} \tag{9}$$

Putting (9) in (3) and rearranging terms, one can get a modified model of inverted pendulum with motor voltage input which is represented using.

$$\frac{K_m}{r}V - \left(\frac{K_m}{r^2R_m} + \frac{b_m}{r^2}\right)\dot{x} = \left(M + m + \frac{J_m}{r^2}\right)\ddot{x} - ml\ddot{\theta} \tag{10}$$

$$-mgl\theta = ml\ddot{x} - (I + ml^2)\ddot{\theta} \tag{11}$$

The equations (10) and (11) can be solved for the $\ddot{\theta}$ and $\ddot{x}$ using Cramer's rule (or any other method) to obtain the state space model of the system [15]. Consider some intermediate variables to make the processes easier. Let,

$$M_1 = \left(M + m + \frac{J_m}{r^2}\right)$$
$$I_1 = ml$$
$$I_2 = (I + ml^2)$$
$$C_1 = \frac{K_m}{r}$$
$$C_2 = \left(\frac{K_m}{r^2R_m} + \frac{b_m}{r^2}\right)$$
$$C_3 = mgl$$

Then the model can be represented as (12),

$$\begin{bmatrix} M_1 & -I_1 \\ I_1 & I_2 \end{bmatrix}\begin{bmatrix} \ddot{x} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} C_1V - C_2\dot{x} \\ -C_3\theta \end{bmatrix} \tag{12}$$

After solving and writing it in state space form the model is represented as shown in (13),

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{M_1C_3}{Z} & 0 & 0 & \frac{I_1C_2}{Z} \\ 0 & 0 & 0 & 1 \\ -\frac{C_3I_1}{Z} & 0 & 0 & \frac{C_2I_2}{Z} \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ -\frac{I_1C_1}{Z} \\ 0 \\ -\frac{C_1I_2}{Z} \end{bmatrix}u(t) \tag{13}$$

Where, $x_1 = \theta, x_2 = \dot{\theta}, x_3 = x, x_4 = \dot{x}, u(t) = V$, and $Z = I_1^2 - M_1I_2$. The parameters for the given system are shown in the table I, those can be estimated using parameter estimation technique.

B. Parameter Estimation

Parameter estimation holds a critical position within the realm of control systems, involving the modeling and simulation of dynamic systems. It encompasses the determination of unknown parameters, relying on observed data or pre-existing knowledge. The accuracy of parameter estimation stands as a cornerstone for the development of reliable control strategies and the creation of precise models for dynamic systems [11,12]. By calibrating model parameters through the application of parameter estimation techniques and subsequently comparing model outputs with experimental or observed data, the precision, and authenticity of the model in capturing system dynamics are considerably enhanced.

Classical control theory or modern tools can be used to estimate the parameters or model of a dynamic system using the output response of the system for a certain input [12,14]. In the MATLAB Parameter Estimation Toolbox, the process of parameter estimation for dynamic systems involves fitting a mathematical model to observed data by adjusting the model parameters. This toolbox offers a range of techniques and functions to facilitate accurate parameter estimation.

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To initiate the process, the user establishes a mathematical model representing the dynamic behavior of the system under investigation. This model should encompass adaptable parameters that require estimation. Subsequently, the observed data, encapsulating the input-output behavior of the system, is prepared for analysis [15]. The toolbox provides the means to configure the estimation algorithm to align with the system's particular requisites and data attributes. This encompasses the selection of the fitting estimation method and specification of any constraints or settings indispensable for the estimation procedure.

With the model and algorithm duly configured, the estimation process is set into motion. The toolbox employs optimization-driven methods or system identification techniques to fine-tune the model's parameters and minimize discrepancies between the model's outputs and the observed data [16]. This iterative journey persists until a contenting alignment is achieved. Following the estimation procedure, the outcomes undergo analysis to gauge the quality and dependability of the estimated parameters. This evaluation typically encompasses an assessment of the goodness of fit, statistical significance, and other pertinent metrics, aimed at corroborating the accuracy of the estimation [15,18].

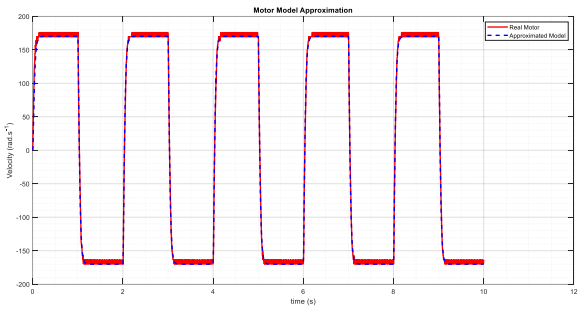

Fig. 3. Comparison of real and estimated motor response for square input signal (voltage or direction) with constant input voltage.

Figure 3 shows the comparison of the response of two dynamic models, The one shown in the red bold line is a real system that represents the real DC motor response for a square input (voltage) signal and the dashed blue line represents the output of the estimated model. It can easily be seen that the estimated motor response perfectly aligns with the real motor response. The values of coefficients of the DC motor and inverted pendulum and cart system are shown in table I.

TABLE I. INVERTED PENDULUM AND CART MODEL PARAMETERS

Parameter	Value	Units
M	0.8530	Kg
m	0.137	Kg
I	0.0014	$Kg.m^2$
l	0.175	m
J_m	2.85×10^{-5}	$Kg.m^2$
K_m	0.0659	$N.m.A^{-1}$
L_m	0.0630	H
R_m	6.7	Ω
B_m	4.55×10^{-6}	$N.m.rad^{-1}.s$
r	0.005	m

C. Nonlinearities and Unmodeled Dynamics

The model of the system exhibits certain non-linearities and unmodeled dynamics, which can introduce challenges and require careful consideration when designing control algorithms [22]. Some of the non-linearities or unmodeled dynamics encountered in this model are as follows:

- Friction: Although our model assumes frictionless bearings, in reality, complete compensation of friction using bearings is not achievable. Therefore, it is necessary to include the effects of friction in the system model. However, incorporating friction introduces complexity, making control more challenging [20].
- Parameter Estimation for Motors: Parameter estimation algorithms are employed to calculate or approximate the motor parameters. However, the estimated parameters may not achieve complete accuracy, and disparities can arise between the genuine motor output and the approximated motor output [21].
- Motor and Motor Driver: The motor possesses defined voltage limitations. The motor ceases operation if the voltage falls below a threshold (e.g., 4V), and the upper bound for permissible motor voltage must not surpass a designated value (e.g., 17V). Crossing the maximum voltage limit could lead to heightened current consumption and the potential risk of motor driver overheating, as it adheres to a power capacity restriction (e.g., 25W).
- Sensor Noise: The sensor implemented within the system introduces noise arising from quantization error. In scenarios involving an encoder, for example, the sensor is confined to offering a finite resolution (e.g., 200 pulses per revolution - PPR), which induces quantization errors.

All these uncertainties and unmodeled dynamics should be accounted while designing the control. By accounting for these factors, one can enhance control performance, ensure system stability, and achieve desired control objectives in the presence of these challenging aspects [16].

D. ABC-Based PID

In this study, the aforementioned ABC algorithm has been employed to validate the PID coefficients for addressing the tracking control issue in the Inverted pendulum and cart system with a DC motor. In this context, the subsequent cost function depicted in equation (11) has been taken into account.

$$J = [W_1 \quad W_2 \quad W_3 \quad W_4 \quad W_5] \begin{bmatrix} T_s \\ M_o \\ e_{ss} \\ u(t) \\ \sum |t \cdot e(t)| \end{bmatrix} \tag{11}$$

Where W_n is a row vector that can be assigned manually in the algorithm to specify the importance of a specific entity in the fitness value. T_s is the settling time, M_o is the overshoot, e_{ss} is the setting error, $u(t)$ is the input signal of the plant, $e(t)$ is the different between reference and output signal mathematically can be written as $e(t) = r(t) - y(t)$.

The inclusion of weights within the fitness function, used to compute fitness values, confers significant advantages in the tuning of PID. Through the allocation of distinct weights to various performance criteria such as settling time, error, or other metrics specific to the system, we adeptly influence their relative importance during the optimization process.

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IV. Results and Discussions

The results of the closed-loop system are depicted in figure 4. It is evident from the system's output that it settles accurately to the reference signal, which was a unit step signal. The settling time of the system is satisfactory, and additional parameters can be observed in table III. In the simulation, both classical (Ziegler-Nichols) and ABC methods were employed for PID tuning in the inverted pendulum and cart system. The corresponding values for these approaches are presented in table II.

TABLE II. PID Parameters comparison

Parameter	Classical	ABC
Kp	-86.8354	-102.3727
Ki	-129.7377	-129.0782
Kd	-11.4289	-5.7026

The control parameters, including settling time, overshoot, and others, can be observed in table III. It is evident that the classical control parameters outperform than those of the ABC algorithm, except for the fitness value. But at this point, it's not good to give any statement in favor of any of the controllers, let's analyze outputs deeply.

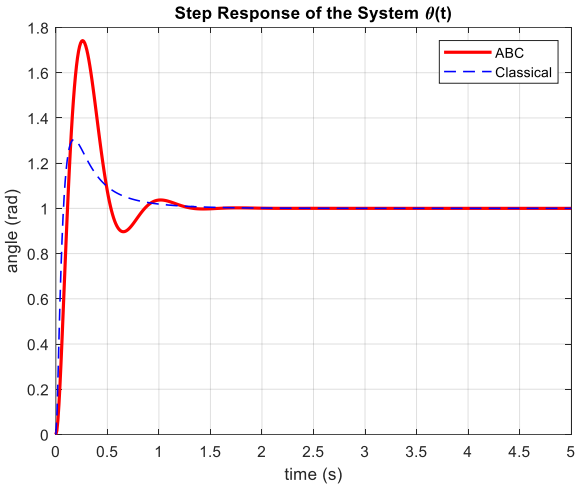

Fig. 4. Comparison of ABC and classically tuned PID control (Step Response)

The output signal illustrated in Figure 5 portrays the motor torque necessary to achieve the response showcased in Figure 4. Evidently, the highest observed torque is 2.16 Nm in the case of the ABC-based PID approach. This observation leads to the conclusion that a motor capable of generating a torque of 2.16 Nm at the rated speed is requisite for this specific system. While the speed can be derived from the transfer function of the third state, as presented in this paper, it does not constitute the primary focus of this study. Interested readers can refer to the GitHub repository (link provided at the end)

The juxtaposition of the two controllers (the classical PID and the ABC PID) yields intriguing insights into their respective performance. On initial inspection, the classical PID controller seems to exhibit a more favorable step response compared to the ABC PID controller, as indicated by the values in Table 2. However, upon closer examination, a notable drawback emerges when analyzing the output response of the classical PID controller, represented by the green curve in Figure 5.

In Figure 5, it is evident that the classical PID controller necessitates an input torque of around 12 Nm to attain the desired step response. This value is significantly higher (roughly 5.7 times) than that required by the ABC PID controller. Such a substantial torque demand presents a pragmatic challenge, as procuring a motor capable of producing such torque might prove intricate or even infeasible within the system's design constraints. Consequently, the ostensibly superior step response performance of the classical PID controller is overshadowed by this limitation.

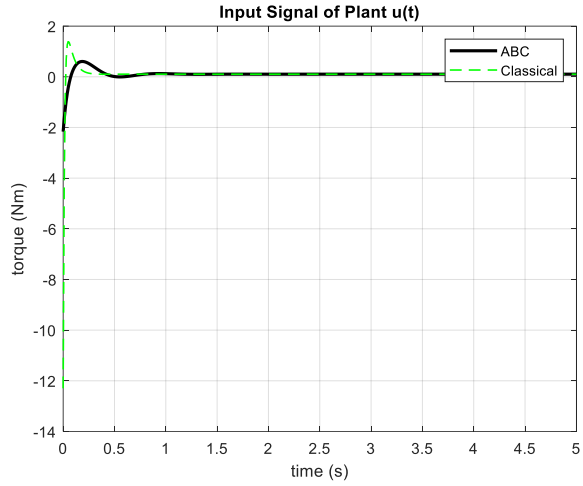

Fig. 5. Comparison of ABC and classically tuned PID control (Input Signal)

On the other hand, the ABC-based PID controller, despite exhibiting slightly inferior step response characteristics compared to the classical PID controller, offers distinct advantages in terms of system feasibility. The oscillations and overshoot observed in the response of the ABC-based PID controller, as depicted in figure 4, can be attributed to the flexibility or reduced stiffness of the inverted pendulum and cart model. While the classical PID controller's stiff response may initially seem desirable, it can potentially cause severe damage to the system, making it impractical for real-world implementation. The ABC-based PID controller, with its more realistic oscillatory behavior, accounts for the system's inherent flexibility and provides a safer and more practical control solution.

TABLE III. Output results comparison

Parameter	Classical	ABC
Settling time	0.9814	1.1908
Overshoot	30.4197	71.3679
Stead state error	0	0
fitness	128.1254	37.2955

Therefore, in this particular scenario, the superiority of the classical PID controller in terms of step response is outweighed by the design constraints and the impracticality of achieving the required torque. The ABC-based PID control approach emerges as the more suitable and practical choice for real-world implementation. By striking a balance between performance and system constraints, the ABC-based PID controller offers a more feasible and effective control solution for the inverted pendulum and cart system.

The fitness curve displayed in figure 6 illustrates the performance of the best solution obtained by the ABC

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algorithm in each iteration. A notable trend is observed as the fitness values consistently decrease over time, disregarding the occasional peaks. This decreasing trend is a desirable characteristic of the ABC algorithm, as it indicates the algorithm's ability to continuously improve and converge towards better solutions. Intermittent peaks in the fitness curve indicate perturbations that prevent stagnation and promote exploration. The algorithm's iterative process consistently decreases fitness values, adapting and refining solutions over time. This dynamic behavior ensures effective optimization and high-quality solutions for the system at hand.

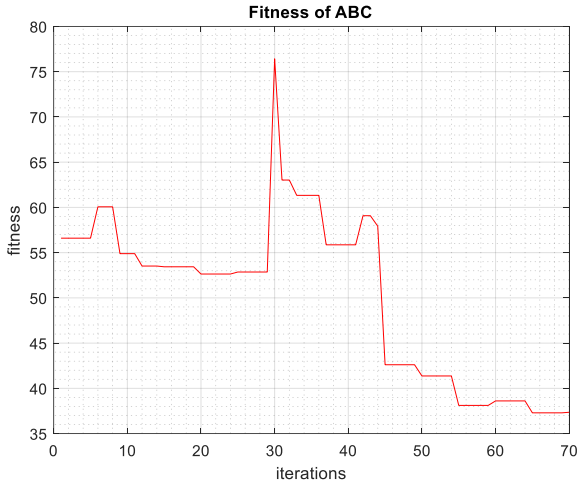

Fig. 6. Fitness vs. Iterations curve of ABC algorithm

V. Conclusion

This paper presented a comparative study between the classical (Zigler-Nicholas) PID tuning method and the evolutionary algorithm, specifically the Artificial Bee Colony (ABC) method, for controlling the Inverted Pendulum and Cart system with an integrated motor. The simulation results revealed that the step response of the classical PID controller exhibited superior performance compared to the ABC-based controller. However, a critical limitation of the classical controller was the excessively high input signal (motor torque) required and the stiffness of the controller (as depicted in figure 4, which posed physical constraints and rendered the solution impractical. To address this challenge, the proposed solution involved reducing the design requirements and re-evaluating the system with lower design constraints. By employing the ABC-based PID controller, a physically viable and optimal solution was obtained. The ABC algorithm demonstrated its capability to provide a low-input solution, achieve excellent settling time, and exhibit desirable stiffness that was practically feasible. Consequently, the ABC based PID controller outperformed the classical PID controller by offering a superior response that adhered to physical limitations. Through this comparative analysis, it became evident that the evolutionary ABC algorithm presented a valuable approach for PID tuning in control systems. By leveraging the advantages of the ABC algorithm, such as its ability to explore the parameter space and adaptively adjust the gains based on performance criteria, engineers can overcome the limitations of traditional PID tuning methods. The ABC-based approach offers the benefits of improved control performance, reduced reliance on manual trial-and-error tuning, increased automation, and enhanced robustness in the presence of system complexities.

Repository: https://github.com/sabirhusnain577/ABC-PID.git

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